Elementary Analysis

Chapter 1, Section 6

A Development of the Real Numbers
Exercises 6.1–6.5, with every subpart

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 1. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, letters in Section 3 denote real numbers. The empty sum is 0, and $0! = 1$.

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6 A development of the real numbers

This section changes the point of view. A real number is to be constructed from rational numbers, rather than assumed in advance. Accordingly, *all elements of cuts and all auxiliary numbers in the construction below are rational.* We use the ordered-field rules for $\mathbb{Q}$, the Archimedean property for $\mathbb{Q}$, and elementary properties of integers. We do not assume that an irrational square root or cube root already exists in order to prove that a set is a cut.

The definition and two preliminary facts

A subset α of $\mathbb{Q}$ is a *Dedekind cut* when:

(i) α is nonempty and is not all of $\mathbb{Q}$;

(ii) it is downward closed: if $a \in \alpha$, $q \in \mathbb{Q}$, and $q < a$, then $q \in \alpha$;

(iii) it has no greatest element: for each $a \in \alpha$, there is $a' \in \alpha$ with $a < a'$.

An *outside element* is a rational number not belonging to the cut. The cut associated with a rational s is

$$s^* = \{q \in \mathbb{Q} : q < s\}.$$

For example, 0^* consists of the negative rational numbers. Its elements are numbers, whereas 0^* itself is a set of numbers.

Preliminary fact 1: an inside element is below every outside element.

Let $a \in \alpha$ and $b \in \mathbb{Q} \setminus \alpha$. Equality $a = b$ is impossible because one number cannot both belong and not belong to the same set. If $b < a$, downward closure would put b in α, also a contradiction. The order of $\mathbb{Q}$ gives

$$a < b.$$

Consequently, if $b \notin \alpha$ and $b \le c$ with $c \in \mathbb{Q}$, then $c \notin \alpha$: otherwise either $c = b$ contradicts the choice of b, or $b < c$ and downward closure puts b in α.

Preliminary fact 2: rational cuts satisfy the definition.

Fix $s \in \mathbb{Q}$. The number $s - 1$ belongs to s^*, so s^* is nonempty. The rational number s does not belong to s^*, so $s^* \neq \mathbb{Q}$. If $q < r$ and $r < s$, transitivity gives $q < s$, proving downward closure. Finally, if $q < s$, the rational midpoint

$$q' = \frac{q+s}{2}$$

satisfies

$$q' - q = \frac{s-q}{2} > 0, \qquad s - q' = \frac{s-q}{2} > 0.$$

Thus $q < q' < s$, and s^* has no greatest element.

6.1 Exercise 6.1: Order, equality, and addition for rational cuts

What we must show. For rational s, t, prove

$$s \le t \iff s^* \subseteq t^*, \qquad s = t \iff s^* = t^*, \qquad (s+t)^* = s^* + t^*.$$

Here the addition on the right of the last equality means

$$s^* + t^* = \{r + u : r \in s^*,\ u \in t^*\},$$

an operation on sets.

Part (a), Step 1: turn an inequality into an inclusion.

Assume $s \leq t$. Take any $q \in s^*$. The definition gives $q < s$. Combining $q < s$ with $s \leq t$ gives $q < t$. Therefore $q \in t^*$. This works for every $q \in s^*$, so $s^* \subseteq t^*$.

Part (a), Step 2: prove the converse by testing a midpoint.

Assume $s^* \subseteq t^*$. Suppose, for a contradiction, that $s > t$. Set $q = (s + t)/2$. This is rational, and

$$q - t = \frac{s - t}{2} > 0, \qquad s - q = \frac{s - t}{2} > 0.$$

Hence $t < q < s$. The inequality $q < s$ puts q in s^*, but $q > t$ prevents q from belonging to t^*. This contradicts the inclusion. Thus $s > t$ is impossible, and the total order gives $s \leq t$.

Conclusion. Both directions have been proved:

$$\boxed{s \leq t \iff s^* \subseteq t^*.}$$

Part (b), Step 1: equal rational numbers give the same set.

If $s = t$, the conditions $q < s$ and $q < t$ are identical. They select the same rational numbers, so $s^* = t^*$.

Part (b), Step 2: the same set determines only one rational endpoint.

If $s^* = t^*$, both $s^* \subseteq t^*$ and $t^* \subseteq s^*$ hold. Part (a), applied in both orders, gives $s \leq t$ and $t \leq s$. Antisymmetry gives $s = t$.

Conclusion. The rational-to-cut correspondence is injective:

$$\boxed{s = t \iff s^* = t^*.}$$

Part (c), Step 1: prove the sum is contained in the target cut.

Let $q \in s^* + t^*$. There are rational $r < s$ and $u < t$ with $q = r + u$. Adding u to the first inequality and s to the second gives

$$q = r + u < s + u < s + t.$$

Consequently $q \in (s + t)^*$.

Part (c), Step 2: split a number below the target endpoint.

Conversely, take $q \in (s + t)^*$. Then $q < s + t$, so

$$\delta = \frac{s + t - q}{2}$$

is a positive rational number. Put $r = s - \delta$ and $u = t - \delta$. Since $\delta > 0$, we have $r < s$ and $u < t$, hence $r \in s^*$ and $u \in t^*$. Moreover,

$$r + u = s + t - 2\delta = s + t - (s + t - q) = q.$$

Thus $q \in s^* + t^*$.

Conclusion. The two inclusions establish

$$\boxed{(s+t)^* = s^* + t^*.}$$

Ordinary rational addition agrees with addition of rational cuts.

6.2 Exercise 6.2: Why the sum of two cuts is another cut

What we must show. Given Dedekind cuts α, β, verify every defining condition for

$$\gamma = \alpha + \beta = \{a + b : a \in \alpha,\ b \in \beta\}.$$

All such sums are rational because $\mathbb{Q}$ is closed under addition.

Step 1: verify that the sum is nonempty.

Choose $a_0 \in \alpha$ and $b_0 \in \beta$, possible because both cuts are nonempty. Then $a_0 + b_0 \in \gamma$, so γ is nonempty.

Step 2: exhibit a rational missing from the sum.

Choose rational outside elements $A \notin \alpha$ and $B \notin \beta$. Preliminary fact 1 gives

$$a < A \quad (a \in \alpha), \qquad b < B \quad (b \in \beta).$$

Adding gives $a + b < A + B$ for every member $a + b$ of γ. In particular, the rational $A + B$ cannot belong to γ. Therefore $\gamma \neq \mathbb{Q}$.

Step 3: prove downward closure by changing one summand.

Let $x \in \gamma$ and let $y \in \mathbb{Q}$ satisfy $y < x$. Write $x = a + b$ with $a \in \alpha$, $b \in \beta$, and define

$$a' = a + (y - x).$$

Since $y - x < 0$, the rational number a' is less than a. Downward closure of α gives $a' \in \alpha$. Also

$$a' + b = a + (y - x) + b = (a + b) + (y - x) = x + y - x = y.$$

Thus $y \in \gamma$.

Step 4: prove that no member is greatest.

Take $x = a + b \in \gamma$. Choose $a' \in \alpha$ with $a' > a$, using the no-greatest-element property of α. Then

$$x' = a' + b \in \gamma, \qquad x' - x = a' - a > 0.$$

Every member x therefore has a larger member x'.

Conclusion. The set $\alpha + \beta$ is nonempty, proper, downward closed, and has no greatest element. Hence

$$\boxed{\alpha + \beta \text{ is a Dedekind cut.}}$$

6.3 Exercise 6.3: The zero cut and a correctly defined negative cut

What we must show. Prove (a) $\alpha + 0^* = \alpha$, and (b) define a cut $-\alpha$ and prove $\alpha + (-\alpha) = 0^*$.

Part (a), Step 1: prove the forward inclusion.

If $x \in \alpha + 0^*$, write $x = a + z$ with $a \in \alpha$ and $z < 0$. Then $x < a$, and downward closure gives $x \in \alpha$. Thus $\alpha + 0^* \subseteq \alpha$.

Part (a), Step 2: represent every original member as a sum.

Take $x \in \alpha$. Choose $a \in \alpha$ with $x < a$ and put $z = x - a$. Then $z \in \mathbb{Q}$, $z < 0$, and $a + z = x$. Hence $z \in 0^*$ and $x \in \alpha + 0^*$.

Conclusion. The two inclusions give

$$\boxed{\alpha + 0^* = \alpha.}$$

Part (b), Step 1: define the negative with a strict endpoint.

Set

$$\boxed{-\alpha = \{q \in \mathbb{Q} : \text{there exists } r \in \mathbb{Q} \setminus \alpha \text{ with } q < -r\}.} \tag{6.3.1}$$

This includes rational numbers strictly below negatives of outside elements.

The strict inequality is necessary. If $\alpha = 0^*$, simply negating every member of its complement would give $\{q \in \mathbb{Q} : q \leq 0\}$. That set has the greatest member 0, and hence is not a cut. Definition (6.3.1) avoids including such a boundary point.

Part (b), Step 2: prove nonemptiness.

Choose $r_0 \notin \alpha$. The rational $q_0 = -r_0 - 1$ satisfies $q_0 < -r_0$, so $q_0 \in -\alpha$.

Part (b), Step 3: prove properness.

Choose $a_0 \in \alpha$. For every rational $r \notin \alpha$, preliminary fact 1 gives $a_0 < r$, hence $-r < -a_0$. Therefore $-a_0 < -r$ cannot hold for any outside element r. Definition (6.3.1) shows $-a_0 \notin -\alpha$, so $-\alpha \neq \mathbb{Q}$.

Part (b), Step 4: prove downward closure.

If $q \in -\alpha$, choose $r \notin \alpha$ with $q < -r$. For any rational $p < q$, we have $p < q < -r$, so the same r witnesses $p \in -\alpha$.

Part (b), Step 5: prove there is no greatest member.

Let $q \in -\alpha$, with outside witness r such that $q < -r$. The rational midpoint $q' = (q - r)/2$ satisfies

$$q' - q = \frac{-r - q}{2} > 0, \qquad -r - q' = \frac{-r - q}{2} > 0.$$

Thus $q < q' < -r$, and $q' \in -\alpha$. Steps 2–5 prove that the proposed negative is a cut.

Part (b), Step 6: establish a rational small-gap lemma.

We need to find an inside point close to an outside point without assuming that a real endpoint already exists.

Claim. For every positive rational h, there is $a \in \alpha$ such that $a + h \notin \alpha$.

Choose $a_0 \in \alpha$, $b_0 \notin \alpha$. Preliminary fact 1 gives $b_0 - a_0 > 0$. The rational Archimedean property supplies $N \in \mathbb{N}$ with

$$Nh > b_0 - a_0.$$

Then $a_0 + Nh > b_0$, so $a_0 + Nh \notin \alpha$ by preliminary fact 1. Consider the finite set

$$J = \{j \in \{1, \dots, N\} : a_0 + jh \notin \alpha\}.$$

It contains N and is nonempty. Let j be its smallest member. The preceding point $a_0 + (j-1)h$ lies inside α: it is a_0 if $j = 1$, and otherwise its index is below the first outside index. Taking

$$a = a_0 + (j-1)h$$

gives $a \in \alpha$ and $a + h = a_0 + jh \notin \alpha$, as claimed. Only the rational Archimedean property and a finite least-index choice were used.

Part (b), Step 7: show that every sum lies in the zero cut.

Take $x = a + q \in \alpha + (-\alpha)$ with $a \in \alpha$, $q \in -\alpha$. Choose $r \notin \alpha$ with $q < -r$. Since $a < r$,

$$x = a + q < a - r < 0.$$

Hence $x \in 0^*$, proving $\alpha + (-\alpha) \subseteq 0^*$.

Part (b), Step 8: represent every negative rational as a sum.

Take $z \in 0^*$, so $z < 0$, and set $h = -z/2 > 0$. The small-gap lemma supplies $a \in \alpha$ with $r := a + h \notin \alpha$. Put $q = z - a$. Then

$$q + r = (z - a) + (a + h) = z + h = z - \frac{z}{2} = \frac{z}{2} < 0.$$

Thus $q < -r$. The outside witness r proves $q \in -\alpha$. Finally, $a + q = a + (z - a) = z$, so $z \in \alpha + (-\alpha)$.

Conclusion. Definition (6.3.1) gives a Dedekind cut satisfying

$$\boxed{\alpha + (-\alpha) = 0^*.}$$

It constructs the additive inverse required by A4 using rational numbers alone.

As a check, when $\alpha = s^*$, every outside element satisfies $r \geq s$. If $q < -r$, then $q < -s$, so $-\alpha \subseteq (-s)^*$. Conversely, if $q < -s$, the outside element $r = s$ witnesses $q \in -\alpha$. Hence $-(s^*) = (-s)^*$.

6.4 Exercise 6.4: Why the proposed product of cuts fails

What we must show. Examine the proposed rule

$$\alpha \cdot_{\text{naive}} \beta = \{ab : a \in \alpha,\ b \in \beta\}$$

using 0^*, 1^*, and $(-1)^*$, and explain why it cannot define multiplication of real numbers.

Part (a), Step 1: identify the three input sets.

Write

$$A = 0^* = \{q \in \mathbb{Q} : q < 0\}, \quad B = 1^* = \{q \in \mathbb{Q} : q < 1\}, \quad C = (-1)^* = \{q \in \mathbb{Q} : q < -1\}.$$

In this exercise, a product of letters means the proposed set of pairwise products. Rational multiplication is commutative, so $AB = BA$, $AC = CA$, and $BC = CB$. Six calculations suffice.

Part (a), Step 2: calculate AA.

Two negative rationals have a positive product, so $AA \subseteq \{q \in \mathbb{Q} : q > 0\}$. Conversely, for any rational $q > 0$,

$$q = (-1)(-q),$$

and both factors belong to A. Therefore

$$AA = \{q \in \mathbb{Q} : q > 0\}.$$

Part (a), Step 3: calculate AC.

A member of A and a member of C are both negative, so their product is positive. For a rational $q > 0$, write

$$q = \left(-\frac{q}{2}\right)(-2),$$

where $-q/2 \in A$ and $-2 \in C$. Consequently

$$AC = \{q \in \mathbb{Q} : q > 0\}.$$

Part (a), Step 4: calculate CC.

Members of C have the forms $-u, -v$ with $u, v > 1$. Their product is $uv > 1$. Hence $CC \subseteq \{q \in \mathbb{Q} : q > 1\}$.

For the reverse inclusion, let $q > 1$ be rational and put

$$r = -\frac{q+1}{2}, \qquad t = -\frac{2q}{q+1}.$$

The denominator $q + 1 > 2$ is nonzero, and $q + 1 > 2$ gives $r < -1$. Since the denominator is positive,

$$\frac{2q}{q+1} > 1 \iff 2q > q + 1 \iff q > 1.$$

Thus $t < -1$. Both factors belong to C, and

$$rt = \frac{q+1}{2}\,\frac{2q}{q+1} = q.$$

We conclude

$$CC = \{q \in \mathbb{Q} : q > 1\}.$$

Part (a), Step 5: calculate AB.

Every product is rational. Conversely, each rational q has a representation with one factor in A and one in B:

- If $q > 0$, use $q = (-q)(-1)$.
- If $q = 0$, use $q = (-1) \cdot 0$; the second factor satisfies $0 < 1$.

- If $q < 0$, use $q = (2q)(1/2)$; the first factor is negative and the second is less than 1.

Therefore $AB = \mathbb{Q}$.

Part (a), Step 6: calculate BB.

Because $q < 0$ implies $q < 1$, we have $A \subseteq B$. Hence

$$\mathbb{Q} = AB \subseteq BB \subseteq \mathbb{Q}.$$

The final inclusion is closure of rational multiplication. Thus $BB = \mathbb{Q}$.

Part (a), Step 7: calculate BC.

Every product again is rational. To represent any rational q:

- For $q > 0$, use $q = (-q/2)(-2)$.
- For $q = 0$, use $q = 0 \cdot (-2)$.
- For $q < 0$, put $u = -q > 0$ and use

$$q = \frac{u}{2(u+1)}\big(-2(u+1)\big).$$

  The second factor is less than -2, so it belongs to C. The first is positive and less than 1 because $2(u+1) - u = u + 2 > 0$. It belongs to B.

Thus $BC = \mathbb{Q}$.

Part (a), Step 8: collect the answers.

Let $\mathbb{Q}_{>0} = \{q \in \mathbb{Q} : q > 0\}$ and $\mathbb{Q}_{>1} = \{q \in \mathbb{Q} : q > 1\}$. All six answers are in the table:

$\cdot_{\text{naive}}$	0^*	1^*	$(-1)^*$
0^*	$\mathbb{Q}_{>0}$	$\mathbb{Q}$	$\mathbb{Q}_{>0}$
1^*	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}$
$(-1)^*$	$\mathbb{Q}_{>0}$	$\mathbb{Q}$	$\mathbb{Q}_{>1}$

Part (b), Step 1: none of these outputs is a cut.

The set $\mathbb{Q}$ is not a cut because it is not a proper subset of itself. The set $\mathbb{Q}_{>0}$ is not downward closed: it contains 1 but not the smaller rational 0. The set $\mathbb{Q}_{>1}$ is not downward closed either: it contains 2 but not the smaller rational 0.

Part (b), Step 2: rational multiplication would give different outputs.

A multiplication consistent with the rational correspondence would require

$$0^* \cdot 0^* = (0 \cdot 0)^* = 0^*.$$

Instead, the proposed rule gives the positive rationals, which are disjoint from 0^*. Likewise, it should give

$$(-1)^* \cdot (-1)^* = 1^*,$$

but instead gives $\mathbb{Q}_{>1}$. For example, $0 \in 1^*$ but $0 \notin \mathbb{Q}_{>1}$.

Conclusion. The proposed pairwise-product rule is not a binary operation on the Dedekind cuts and does not reproduce rational multiplication. A valid definition has to account for signs and downward closure.

6.5 Exercise 6.5: Constructing irrational cuts using rational inequalities

What we must show. Prove all three claims:

(a) $\alpha = \{q \in \mathbb{Q} : q^3 < 2\}$ is a cut, but $\{q \in \mathbb{Q} : q^2 < 2\}$ is not;

(b) α is not the cut of any rational number;

(c)
$$\beta = 0^* \cup \{q \in \mathbb{Q} : q \geq 0 \text{ and } q^2 < 2\}$$
is a cut and is not the cut of any rational number.

Plan. Rational polynomial inequalities will verify the cut conditions. Explicit positive rational increments will rule out a greatest element and will rule out rational endpoints. No pre-existing irrational root is needed.

Preliminary calculation 1: cubes preserve strict order.

For rational $u < v$,
$$v^3 - u^3 = (v - u)(v^2 + uv + u^2).$$
The first factor is positive. The other factor satisfies
$$v^2 + uv + u^2 = \left(v + \frac{u}{2}\right)^2 + \frac{3u^2}{4}.$$
If $u \neq 0$, the last term is positive and the square is nonnegative. If $u = 0$, then $v > 0$ and the factor equals $v^2 > 0$. Thus $v^3 - u^3 > 0$ for all rational signs, proving $u^3 < v^3$.

Preliminary calculation 2: bound explicit small increments.

Fix $s \in \mathbb{Q}$, let h be rational with $0 < h \leq 1$, and put
$$C_3(s) = 3s^2 + 3\,|s| + 1 > 0.$$
The absolute value is rational because it is either s or $-s$. Multiplying $h \leq 1$ by $h > 0$ gives $h^2 \leq h$; multiplying again gives $h^3 \leq h^2 \leq h$. Expansion yields
$$\begin{aligned}(s+h)^3 - s^3 &= 3s^2h + 3sh^2 + h^3 \\ &\leq 3s^2h + 3\,|s|\,h^2 + h^3 \\ &\leq (3s^2 + 3\,|s| + 1)h = C_3(s)h.\end{aligned}$$
Using $-s \leq |s|$ gives the corresponding decrement bound:
$$\begin{aligned}s^3 - (s-h)^3 &= 3s^2h - 3sh^2 + h^3 \\ &\leq 3s^2h + 3\,|s|\,h^2 + h^3 \\ &\leq C_3(s)h.\end{aligned}$$
The bounds hold even when s is negative or zero.

For squares, define the positive rational $C_2(s) = 2\,|s| + 1$. Expansion gives
$$(s+h)^2 - s^2 = 2sh + h^2 \leq C_2(s)h,$$

$$s^2 - (s-h)^2 = 2sh - h^2 \le 2|s|\,h \le C_2(s)h.$$

Every denominator $2C_3(s)$ and $2C_2(s)$ below is positive.

Part (a), Step 1: prove that the cubic set is nonempty and proper.

The rational 0 lies in α, since $0^3 = 0 < 2$. The rational 2 does not lie in α, since $2^3 = 8 > 2$. Hence α is a nonempty proper subset of $\mathbb{Q}$.

Part (a), Step 2: prove downward closure of the cubic set.

Let $q \in \alpha$ and let rational $p < q$. Strict increase of cubes gives

$$p^3 < q^3 < 2.$$

Thus $p \in \alpha$. This argument includes negative p and q.

Part (a), Step 3: construct a larger cubic-set member.

Let $q \in \alpha$, so the rational $\delta = 2 - q^3$ is positive. Choose

$$h = \min\left\{\frac{1}{2}, \frac{\delta}{2C_3(q)}\right\}.$$

Both entries of the minimum are positive rational numbers. Therefore h is rational, $0 < h \le 1/2 < 1$, and $C_3(q)h \le \delta/2$. The increment bound gives

$$(q+h)^3 \le q^3 + C_3(q)h \le q^3 + \frac{\delta}{2} < q^3 + \delta = 2.$$

Hence $q + h \in \alpha$ and $q + h > q$. Thus α has no greatest member.

Conclusion. The cubic set $\{q \in \mathbb{Q} : q^3 < 2\}$ satisfies all defining conditions and is a Dedekind cut.

Part (a), Step 4: test the unmodified square condition.

Let $E = \{q \in \mathbb{Q} : q^2 < 2\}$. It contains 0, since $0^2 < 2$. But $-2 < 0$ and $(-2)^2 = 4 > 2$, so $-2 \notin E$. This contradicts the downward closure that any cut containing 0 would have to satisfy.

Conclusion. The set $\{q \in \mathbb{Q} : q^2 < 2\}$ is not a cut. Its exclusion of sufficiently negative rationals prevents downward closure.

Part (b), Step 1: suppose the cubic cut has a rational endpoint.

Suppose $\alpha = s^*$ for some $s \in \mathbb{Q}$. If $s^3 < 2$, then $s \in \alpha$. But $s \notin s^*$ because $s < s$ is false. This contradicts equality of the two sets. Thus $s^3 < 2$ is impossible.

Part (b), Step 2: exclude an endpoint whose cube is too large.

If $s^3 > 2$, put $\delta = s^3 - 2 > 0$ and choose

$$h = \min\left\{\frac{1}{2}, \frac{\delta}{2C_3(s)}\right\} > 0.$$

The rational $r = s - h$ satisfies $r < s$, so $r \in s^*$. However,

$$s^3 - r^3 \le C_3(s)h \le \frac{\delta}{2}.$$

Rearranging gives

$$r^3 \geq s^3 - \frac{\delta}{2} > s^3 - \delta = 2.$$

Therefore $r \notin \alpha$, another contradiction. The total order excludes both strict alternatives and forces $s^3 = 2$.

Part (b), Step 3: prove that a rational cube cannot equal 2.

Write $s = m/n$ in lowest terms, with $m, n \in \mathbb{Z}$ and $n > 0$. The equation $s^3 = 2$ gives $m^3 = 2n^3$, so m^3 is even. An odd integer has the form $2k + 1$ and has odd cube:

$$(2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1 = 2(4k^3 + 6k^2 + 3k) + 1.$$

Hence m must be even; put $m = 2u$ with $u \in \mathbb{Z}$. Substitution gives

$$8u^3 = 2n^3, \qquad n^3 = 4u^3.$$

Now n^3 is even, so the same parity argument makes n even. Then m and n have the common positive factor 2, contradicting lowest terms.

Conclusion. The cubic cut is not s^* for any rational s. It represents an irrational element of the cut construction, later identified with the positive cube root of 2 after the real-number construction is completed.

Part (c), Step 1: prove nonemptiness and properness.

The corrected set is

$$\beta = \{q \in \mathbb{Q} : q < 0\} \ \cup \ \{q \in \mathbb{Q} : q \geq 0 \text{ and } q^2 < 2\}.$$

The rational 0 belongs to its second part, since $0 \geq 0$ and $0^2 < 2$. Thus β is nonempty. The rational 2 is not negative and its square 4 is not less than 2. It lies in neither part, so $\beta \neq \mathbb{Q}$.

Part (c), Step 2: verify downward closure by separating signs.

Let $q \in \beta$ and let $p \in \mathbb{Q}$ satisfy $p < q$. If $p < 0$, it belongs to $0^* \subseteq \beta$. If $p \geq 0$, then $q > p \geq 0$. Thus q cannot belong to the negative part of β, and its membership must give $q^2 < 2$. Moreover,

$$q^2 - p^2 = (q - p)(q + p) > 0,$$

since $q - p > 0$ and $q + p > 0$. Hence $p^2 < q^2 < 2$. Together with $p \geq 0$, this puts p in the second part of β. Both signs of p therefore give membership.

Part (c), Step 3: find a larger member when the chosen member is negative.

If $q \in \beta$ and $q < 0$, put $q' = q/2$. Then $q' - q = -q/2 > 0$ and $q' < 0$. So $q' > q$ and $q' \in 0^* \subseteq \beta$.

Part (c), Step 4: find a larger member in the nonnegative case.

If $q \in \beta$ and $q \geq 0$, then $\delta = 2 - q^2 > 0$. Choose the rational

$$h = \min\left\{\frac{1}{2}, \frac{\delta}{2C_2(q)}\right\} > 0.$$

It satisfies $h \leq 1$ and $C_2(q)h \leq \delta/2$, so

$$(q + h)^2 \leq q^2 + C_2(q)h \leq q^2 + \frac{\delta}{2} < q^2 + \delta = 2.$$

Because $q + h \geq 0$, it lies in β, and $q + h > q$. This case includes $q = 0$.

Conclusion. The corrected square set β is nonempty, proper, downward closed, and has no greatest rational. It is a Dedekind cut. Including all negative rationals repairs the failure in part (a).

Part (c), Step 5: suppose the corrected square cut is rational.

Assume $\beta = s^*$ for some rational s. The rational 1 belongs to β, because $1 \geq 0$ and $1^2 < 2$. Therefore $1 \in s^*$, which means $1 < s$. In particular, $s > 0$.

If $s^2 < 2$, then $s \geq 0$ and $s^2 < 2$ put s in β. But $s \notin s^*$, contradicting equality. Thus $s^2 < 2$ is impossible.

Part (c), Step 6: exclude an endpoint whose square is too large.

If $s^2 > 2$, let $\delta = s^2 - 2 > 0$ and choose

$$h = \min\left\{\frac{1}{2}, \frac{s}{2}, \frac{\delta}{2C_2(s)}\right\}.$$

All entries are positive rationals. Thus $0 < h \leq 1$, $h \leq s/2$, and $C_2(s)h \leq \delta/2$. The rational $r = s - h$ satisfies

$$r \geq \frac{s}{2} > 0, \qquad r < s.$$

The latter inequality puts r in s^*, but

$$r^2 \geq s^2 - C_2(s)h \geq s^2 - \frac{\delta}{2} > s^2 - \delta = 2.$$

Since $r > 0$ and $r^2 > 2$, it is not in either part of β. This contradicts $\beta = s^*$. Both strict alternatives have been excluded, so equality of these cuts would force $s^2 = 2$.

Part (c), Step 7: exclude a rational square equal to 2.

Write $s = m/n$ in lowest terms, with $m, n \in \mathbb{Z}$ and $n > 0$. Then $s^2 = 2$ implies $m^2 = 2n^2$, so m^2 is even. An odd integer $2k + 1$ has odd square:

$$(2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Hence m is even; write $m = 2u$. Substitution gives

$$4u^2 = 2n^2, \qquad n^2 = 2u^2.$$

Now n^2 is even, so the same argument makes n even. Both numerator and denominator would be divisible by 2, contrary to lowest terms.

Conclusion. The corrected square cut is not the cut of a rational number. It represents the irrational element later identified with the positive square root of 2. Its cut conditions and irrationality were established using rational arithmetic, explicit inequalities, and the cut axioms, without assuming that real square root in advance.